National Chains and Trends in Retail Productivity Dispersion

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Abstract

Productivity dispersion within an industry is an important characteristic of the business environment, potentially reflecting factors such as market structure, production technologies, and reallocation frictions. The retail trade sector saw significant changes between 1987 and 2017, and dispersion statistics can help characterize how it evolved over this period. In this paper, we shed light on this transformation by developing public-use Dispersion Statistics on Productivity (DiSP) data for the retail sector for 1987 through 2017. We find that from 1987 through 2017, dispersion increased between retail stores at the bottom and middle of the productivity distribution. However, when we weight stores by employment dispersion, the middle of the distribution is lower initially and decreases over time. These patterns are consistent with a retail landscape featuring more and more activity taking place in chain stores with similar productivity. Firm-based dispersion measures exhibit a similar pattern. Further investigation reveals that there is substantial heterogeneity in dispersion levels across industries.

Keyword: retail, reallocation, business cycles, productivity dispersion

JEL Classification: D24, E24, L81

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1. Introduction

Within-industry productivity dispersion is an important characteristic of the business environment, and can indicate potential for gains from reallocation, variable markups, or variation in production technologies (see, e.g., Hsieh and Klenow, 2009; Bils et al. 2021; Edmonds, Midrigan, and Xu, 2023; Blackwood, Haltiwanger, and Wolf, 2025). Prior work has focused on measuring dispersion in the manufacturing sector (Cunningham et al., 2023) and the economy as a whole (Decker et al., 2020). This paper focuses on retail trade—a sector that has seen significant changes since the late 1980s.

Within the retail sector, there has been substantial reallocation of activity from small to large firms, even though the retail share of aggregate economic activity has remained constant (Foster, Haltiwanger, Klimek, Krizan, Ohlmacher, 2016). It is not clear, *a priori*, whether this transformation would result in decreased or increased productivity differences across stores. On one hand, stores within an industry may have become more similar over time as national chains have come to dominate many retail industries. If stores within a chain have similar productivity, we would expect within-industry dispersion to decrease. On the other hand, if new stores that are part of chains are more productive than existing stores, then productivity dispersion could increase. Which of these forces dominates is important for understanding the allocation of inputs and the distribution of productivity changes.

To examine this question and others, we develop new, publicly available statistics on within-industry labor productivity dispersion for the retail sector.[1] These statistics, which are derived from the Census of Retail Trade (CRT), include measures of dispersion for the middle of the productivity distribution (the interquartile range or IQR) and the tails. By looking at these different dispersion measures, we can get a clearer picture of the impact of the documented shift

[1] New Dispersion Statistics on Productivity for the retail sector are a joint product of the Bureau of Labor Statistics and the Census Bureau and are available at https://www.census.gov/disp and https://www.bls.gov/productivity/articles-and-research/dispersion-statistics-on-productivity/home.htm.

toward larger stores on productivity distribution in the retail sector. Given the theoretical importance of higher moments such as dispersion in characterizing misallocation, changes in dispersion also have implications for growth and welfare.

We find that, like manufacturing, there is substantial productivity dispersion in both the middle and the tails of the distribution. But in contrast to manufacturing, there is more dispersion in the lower tail of the productivity distribution than in the upper tail, and it matters whether we weight stores by employment. This is because much of the expansion of retail trade was due to national chains opening new stores that are larger than those in single-store firms and like other national chain stores. For similar reasons, firm-based dispersion is greater than store-based dispersion. But that changes when we weight stores and firms by employment. We also find that both firm and establishment IQRs decline when weighted by employment. This suggests that reallocation of economic activity towards national chains has been accompanied by stores and firms being more similar in terms of productivity.

The new dispersion statistics for the retail sector that we present here are designed to complement the official BLS productivity statistics. This is an important contribution to our understanding of the aggregate statistics because decreasing dispersion is potentially related to reallocation.[2] These statistics expand the coverage of the joint BLS-Census Dispersion Statistics on Productivity (DiSP) experimental data product, which—until now—covered only the manufacturing sector.

The remainder of the paper describes how we measure productivity and then documents trends in productivity levels and dispersion. Section 2 describes our data and compares our micro-aggregated statistics to published BLS statistics. Section 3 explores different ways of measuring productivity dispersion across stores in the retail sector. Section 4 compares dispersion at the store and firm level. Section 5 provides a detailed look at five industries that

[2] See Syverson (2004), Decker et al. (2020), Cunningham et al. (2023) for discussions related to manufacturing.

illustrate the heterogeneity across industries. Section 6 presents dispersion measured using payroll as the indicator of labor inputs. Section 7 compares our results for retail trade to analogous publicly available statistics on manufacturing. Section 8 concludes.

2. Measuring Productivity

Our analysis is restricted to labor productivity (LP), because the Census of Retail Trade data lack data on capital and other inputs that are needed to calculate TFP. To calculate labor productivity, we need four things: sales, employment, hours per employee, and industry deflators. In brief, we take sales and employment from the Census microdata, and we use the same values of hours per employee and industry deflators as the BLS. In the next subsection, we provide more details on our method of calculating productivity concepts at the micro level, providing points of comparison to official BLS measures.[3] Then, we take a closer look at comparisons of aggregate measures of productivity in our data to official measures.

2.1 Establishment-level Productivity using Census Data

To measure establishment-level LP, we combine restricted-use establishment-level data from the Census Bureau's Census of Retail Trade (CRT) with public-use industry-level data on prices and hours worked from the BLS. The CRT is conducted every 5 years (those ending in "2" and "7") as part of the Economic Census. It includes all retail establishments of multi-store firms and a sample of single-store firms and collects data on sales, employment, and other relevant information. To increase the coverage of single-unit firms, the Census Bureau uses information from administrative records to impute sales data. We use the Longitudinal Business Database to create longitudinal links and time-series-consistent industry codes at the establishment level (see Chow et al., 2021). Our sample consists of all establishments in the CRT and Census of Wholesalers data files that are assigned a NAICS code in the retail sector.

[3] We refer the reader to official BLS documentation for more information on how the official measures are created. See https://www.bls.gov/opub/hom/inp/home.htm for a full description of how the BLS calculates industry statistics.

Our goal is to construct an output measure from the microdata that matches the BLS measure as closely as possible. The underlying data source for our output measure, sales in the Census of Retail Trade, is the same as the one used by the BLS. When industry codes change, the BLS re-estimates sales by industry for earlier years to create a consistent series over time. In contrast, our data have NAICS codes for each establishment in every year (Chow et al., 2021). These codes are based on more information than is available in the aggregate data. These differences lead our estimate of sales to differ more before the SIC-to-NAICS transition in 2002, and for industries where NAICS codes have changed between 2002 and 2017.

We measure labor input as total hours worked. We calculate total annual hours worked by multiplying total employment by the average annual hours worked per employee in the most detailed NAICS code available from the Division of Industry Productivity at the BLS. Thus, almost all of the between-establishment variation in hours worked in a 4-digit industry is due to variation in employment across establishments in the industry.[4] For establishment *e* in industry *j*, log labor productivity is defined as (time index is dropped for simplicity):

$$LP_{ej} = \ln\left(\frac{real\ sales_e}{hours_j \times employment_e}\right)$$

One challenge with measuring employment inputs for retail establishments is that employment can be seasonal, while CRT employment is measured for the pay period that includes March 12. In addition, part-time work is common in retail, but the CRT does not differentiate between full-time and part-time employees. Because the number of employees in the CRT in March may not reflect each establishment's use of labor inputs throughout the year, we

[4] A very small amount of labor input variation can come from variation in hours worked across 6-digit industries within a 4 digit-industry. When we do weighted versions of our statistics, we use employment weights to aggregate entities within an industry and estimated hours of employment to aggregate across industries. We collectively call these weights "employment weights."

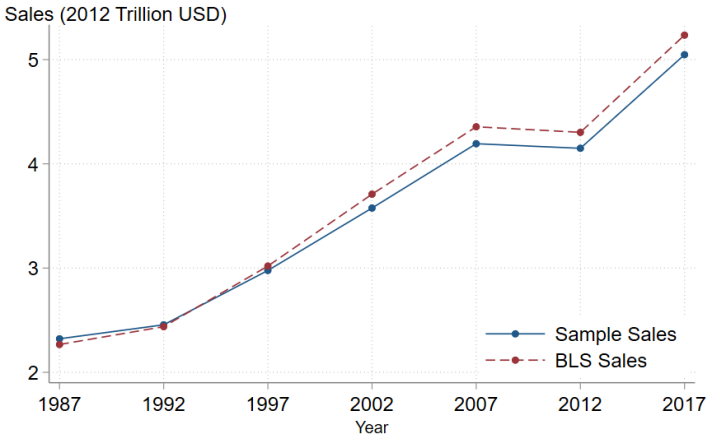
also present estimates using real annual payroll (instead of hours) in the denominator. Payroll-based productivity is potentially a more comprehensive measure of labor for establishments with both full- and part-time workers. However, it is important to keep in mind that observed differences in payroll may reflect unobserved differences in occupational mix, overtime, worker quality, and worker-firm effects.

2.2 Comparing Micro-Aggregated Data to Published Industry Data

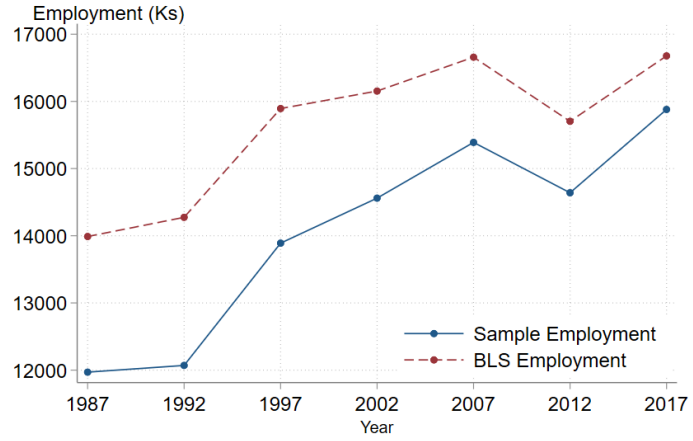
The results in this paper are intended to complement the official BLS industry-level productivity measures, so official measures serve as a useful yardstick for our analysis (Cunningham et al., 2023). BLS official statistics use the Economic Census for sales, as we do, and the same industry deflators and hours per employee measures. Figure 1a shows that our measure of sales closely tracks the BLS measure. Therefore, the primary source of difference between the productivity estimates in our data and the official BLS estimates will come from employment. Our measure is derived from the same Census microdata, as our sales measure, while the BLS relies on employment estimates from the Current Employment Statistics and Current Population Survey. Figure 1b summarizes the difference. Our employment estimates are lower, possibly driven by the fact that the BLS accounts for non-employers. However, this difference decreases over time.

This discrepancy leads our estimates of productivity levels to be more similar to the BLS over time but lowers our estimates of productivity growth relative to the BLS. Figure 1c compares annual average productivity growth rates from our microdata to official BLS growth rates. The two series exhibit nearly identical growth rates over the 1987-1992 period. Starting in the 1992-1997 period, the BLS growth rate is greater. In subsequent periods, two series have similar trends. Despite the differences between the two series, they tell the same qualitative story and suggest that the micro-aggregated productivity estimates are consistent with published BLS data.

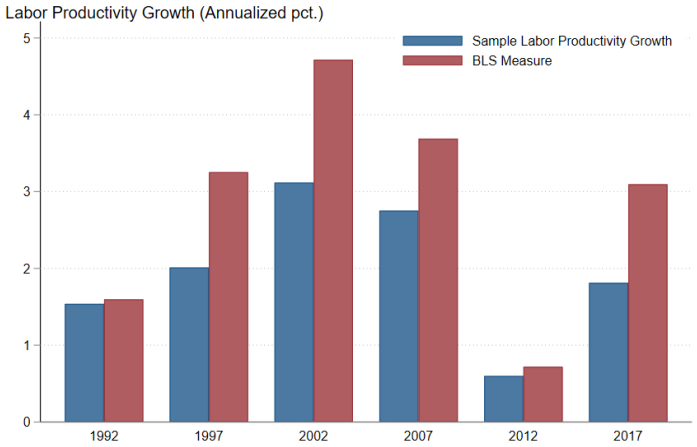
Figure 1: Comparison Between Sample Statistics and BLS Statistics



a. Retail Sales



b. Retail Employment Levels



c. Productivity Levels

Notes: BLS sales and employment use the BLS industry statistics. Sample sales and employment are from the sample created by combining the CRT, CWH, and LBD. Microdata labor productivity growth plots five-year growth rates using sales and employment from the CRT and hours from the BLS industry productivity statistics. Productivity is calculated for each establishment and then averaged to calculate aggregate productivity. The establishments in the top and bottom one percent of the productivity distribution for each industry group are excluded from the calculation. Official BLS Measure is the percent change in index levels.

3. Dispersion Across Establishments

In this section, we explore several aspects of productivity dispersion with the goal of shedding light on the transformation of the retail trade sector that has occurred since 1987. The transformation of the retail sector involved a reallocation of economic activity towards large firms and towards industries that sell multiple products. This has created a more homogeneous shopping experience for consumers in different retail markets (Smith and Ocampo, 2025). In this section, we investigate whether the transformation has also caused retail stores to be more similar in terms of productivity. That is, have the middle and tails of the productivity distribution begun looking more similar as national chains have increased in importance?[5]

3.1 Middle Dispersion Declining, Left Tail Increasing

We start by considering the interquartile range of the productivity distribution. Figure 2a shows that on an unweighted basis the middle of the productivity distribution is essentially unchanged over time. The establishment at the 75th percentile is about 150 percent more productive than the one at the 25th percentile. The right tail (Figure 2b) initially shows a slight increase between 1987 and 1992 before becoming flat for the final 25 years of the study at 100, indicating the 99th percentile store is twice as productive as the one at the 90th percentile in the average industry. The lower tail (Figure 2c) tells a different story with dispersion rising from 100 to over 150 between 1987 and 2017, indicating the difference between the 10th and 1st percentile stores has increased substantially over time.

However, these unweighted measures miss an important aspect of the retail sector. There is substantial variation in industry size and store size within an industry. In particular, stores of national chains are on-average 3 times larger than those of single-units. Giving those stores equal weight in productivity dispersion measures risks missing important aspects of

[5] The IQR is the difference in log labor productivity between establishments at the 75th and 25th percentiles of the distribution, calculated for each industry-year pair. Other percentile ranges are calculated similarly. All percentiles are calculated using an interval method for disclosure avoidance reasons.

reallocation.[6]

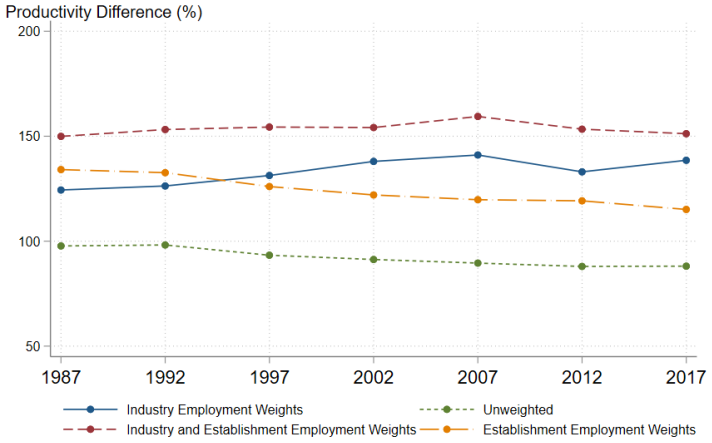
Figures 2a, b, and c also show the three additional combinations of store and industry weighting. Weighting has a minimal impact at the tails of the productivity distribution, possibly because stores at the tails are similar sizes. Weighting by industry share of employment gives substantially more weight to Grocery Stores and General Merchandise Stores, the two largest 4-digit industries. Weighting by industry share of employment lowers IQR dispersion by about 20 percentage points on average but has minimal impact on the time trend.

The more interesting story comes from looking at what happens when stores (or establishments) are weighted according to their employment. We find the IQR decreases by 10 to 20 percentage points over time when we weight in this way (Establishment Employment Weights line in Figure 2a). This result suggests that the reallocation of economic activity towards large stores of national chains has decreased the differences between establishments.

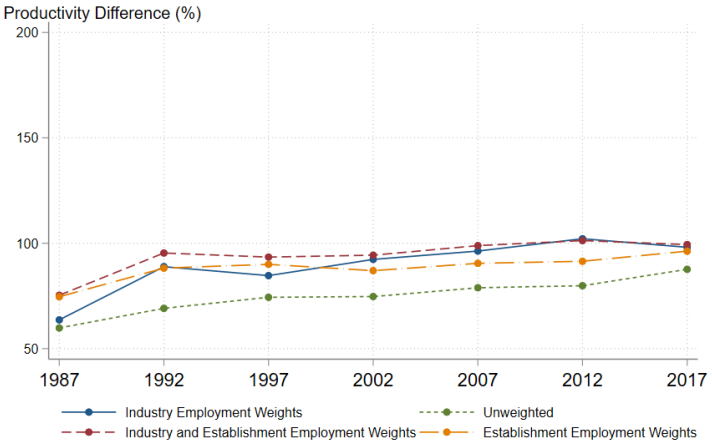
Our results to this point suggest that the shift in retail toward larger establishments has led to more similarity in stores. However, there is still substantial variation in productivity across stores. A 90 percent difference in productivity between the 75th and 25th percentile store is substantial for stores that are presumably selling similar products at a similarly constructed store. These productivity differences can reflect variation in how busy stores are, the size of the stores, or investments in technologies such as self-checkout machines. They could also reflect variation in prices across geographic areas as our deflators are national.

[6] Another way to think about weighted distribution statistics is that they (roughly) measure the dispersion of worker productivity, where each worker in a store is assumed to have the same productivity (that of the store).

Figure 2: Average Productivity Dispersion



a. IQR (75-25 Range)



b. 99-90 Range



c. 10-1 Range

Notes: Graph lines represent the average ranges (IQR, 99-90, 10-1) across industries when using different weighting schemes. Industry Employment Weights combine industries using their estimated share of employment hours. Unweighted is an unweighted arithmetic average. Establishment weights are calculated using each establishment's share of employment within an industry. Figures are based on authors' calculations using CRT and BLS data.

3.2 Dispersion Across Industries

The dispersion results in the previous section tell us about average dispersion across industries but say little about how dispersion varies across industries. If a small number of industries have high dispersion while others have minimal dispersion that could indicate that our results are driven by industry classification; some industry codes are more broadly defined than others. We find that this is not the case. The level of dispersion for even the industry at the 25th percentile of the distribution tends to be quite high and the time trends of industries at the 25th and 75th percentile of the distribution qualitatively follow the trends of the mean and median industry.

Figure 3 shows percentiles of the across-industry distribution of IQRs, 99-90 ranges, and 10-1 ranges. We plot the 25th, 75th, and 50th percentile of the cross-industry distribution in addition to the means that were plotted in Figure 2. We present results with industry and store employment weights and without any weighting in Figure 3.[7] With 27 industries the 25th percentile corresponds to the industry with the 7th highest dispersion in a given year, the 50th percentile corresponds to the 14th, and the 75th corresponds to the 20th.

In panels a and f, we see that a wide range of industries exhibit IQRs over 100. Comparing the two panels, we see that the range between the 75th and 25th percentile is similar for weighted and unweighted dispersion statistics, but the mean IQR is lower when weights are used and their trends differ. The 75th and 25th percentile lines converge when no weights are used (panel a), indicating that industries are looking more similar in terms of dispersion. The convergence is due both a decrease in the 75th percentile IQR and an increase in the 25th percentile IQR. But 75th and 25th percentile trends are roughly parallel in the industry-store weighted graph, even though the mean IQR is decreasing.[8]

[7] Results for the Industry Weighted and Store Weighted methods fall between these two extremes
[8] It is not necessarily the same industries that occupy these percentiles across time and weighting schemes.

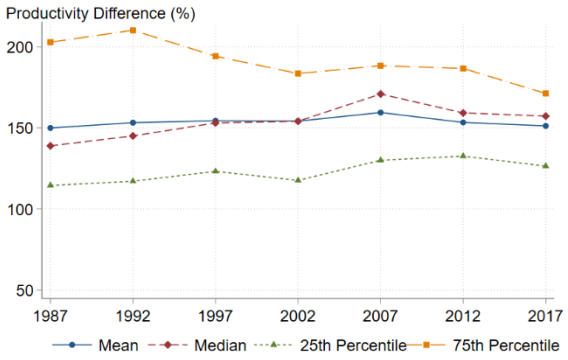
3.3 Tail Dispersion Measures

Turning to the tails of the productivity distribution, weighting continues to have a minimal impact in the 99-90 dispersion statistics. In all cases, there is a slight upward time trend, with the weighted graph having slightly lower levels. These findings are consistent with the reallocation of economic activity towards large firms, mostly impacting the middle of the productivity distribution.

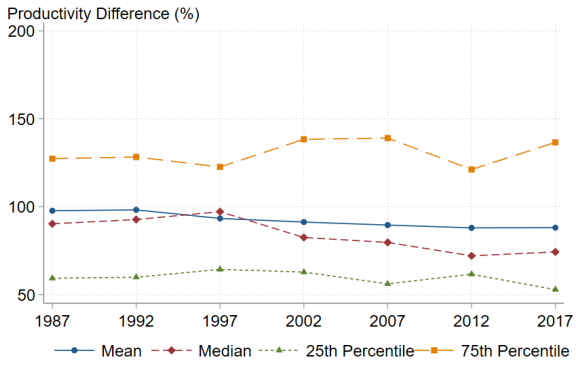
At the other end of the productivity distribution (panels c and f), all percentiles and weighting methods tell the same story of increasing dispersion. On a weighted basis, there is more variation across industries, and the 75th percentile is flat or falling for the final 10 years of the sample. This finding contrasts with manufacturing, where the upper tail exhibits more dispersion. The significant dispersion among the least productive retailers may reflect either persistent distortions that keep unproductive stores afloat or niche shops meeting specific needs that appear relatively unproductive.

Overall, our findings suggest that unweighted establishment-level productivity dispersion is rising in retail, just as it is in the manufacturing sector. However, as activity is increasingly concentrated in large establishments, this has led to declining dispersion when weighting stores by employment. This is most important for central dispersion measures, suggesting more concentration of large establishments in the center of the distribution. In the tails of the distribution, employment-weighting matters less, and dispersion is unambiguously rising (to different degrees) at both extremes. Industry-weighting changes some trends quantitatively, but does not alter the general patterns.

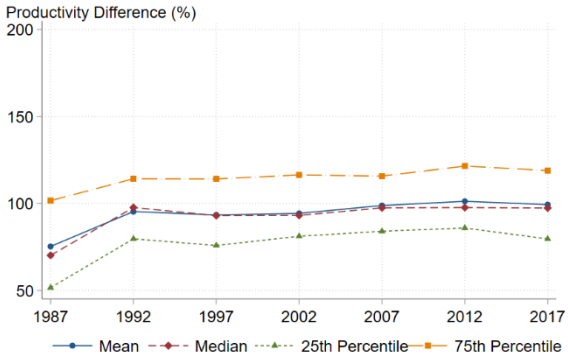
Figure 4: Dispersion across Industries: Unweighted vs. Weighted



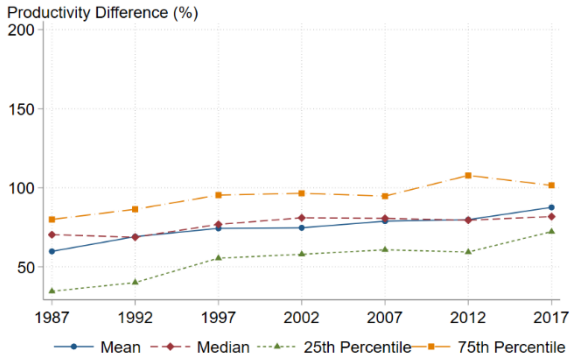
a. Unweighted IQR



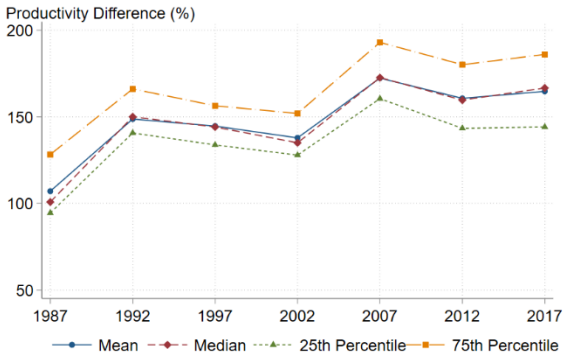
d. Industry-Store Weighted IQR



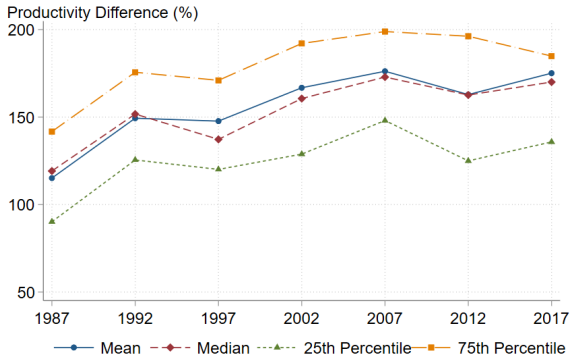
b. Unweighted 99-90 Range



e. Industry-Store Weighted 99-90 Range



c. Unweighted 10-1 Range



f. Industry-Store Weighted 10-1 Range

Notes to Figures 5a-5f: Graph lines represent the variation across industries in the within-industry statistics. The lines indicate the 25th percentile, median, 75th percentile, and mean within-industry IQR across industries. The y-axis is expressed in percent multiples that indicate how much more productive establishments at the top of the range are relative to those at the bottom of the range. For example, a multiple of 100 indicates that stores at the top of the range are twice as productive as stores at the bottom. Figures are based on authors' calculations using CRT and BLS data.

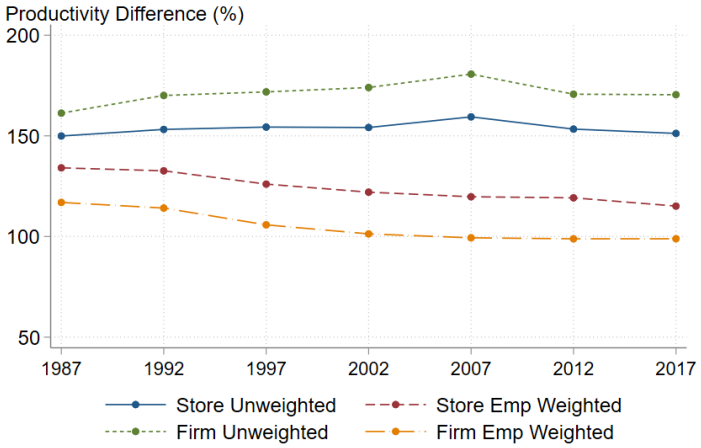
4. Firm-Level Dispersion

To better understand the role of consolidation in retail, we shift our focus from store-level to firm-level statistics. Between 1987 and 2017, retail trade has evolved from a sector of mostly single-store firms to one dominated by national chains. So far, we have seen evidence of this evolution when we compared store-weighted and unweighted dispersion. One aspect of the retail trade sector that is different from other sectors, such as manufacturing, is that firms tend to expand by new stores. Stores opened by national chains tend to be larger than single stores. Much new technology—such as self-checkout, inventory control systems, etc.—can be relatively easily incorporated into existing stores. Thus, we expect much of the variation in productivity across establishments to be due to between-firm rather than within-firm variability. Large firms with many similarly-productive establishments create concentration points in the weighted firm productivity distribution, while large firms have almost no influence on the unweighted distribution because there are more than 500,000 single-unit firms and less than 1000 national firms.

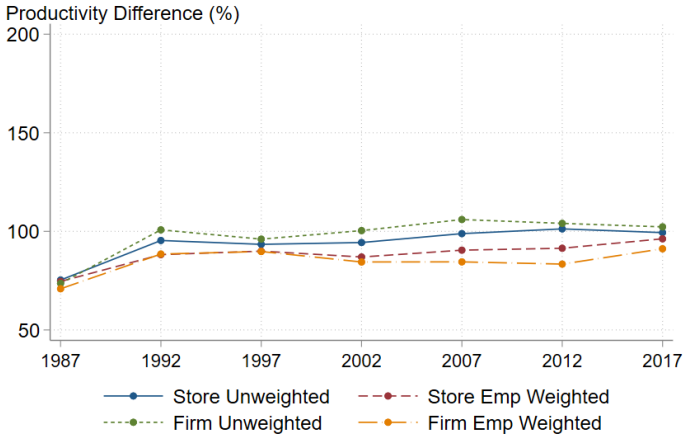
4.1 Results

Figure 5 compares store-level dispersion (from Figure 2) and firm-level dispersion, both hours weighted and unweighted. Starting with panel a, both firm-level and store-level IQRs are higher than their respective weighted IQRs. Unweighted firm-level IQRs are higher than unweighted store-level IQRs, consistent with overall dispersion being driven primarily by differences across firms. This should not be too surprising because the existence of large chains that have many similar stores has the same effect on store-level dispersion as weighting. The relationship between store and firm-level IQRs changes when we weight by hours—firm-level IQRs are lower. As with the store-level dispersion statistics, the downward slopes of the weighted IQRs reflect the shift from single stores to national chains.

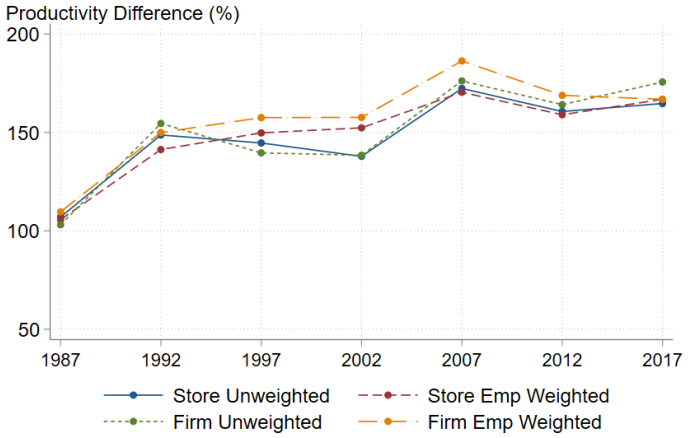
Figure 5: Firms vs. Stores



a. Firm and Store IQRs



b. Firm and Store 99-90 Ranges



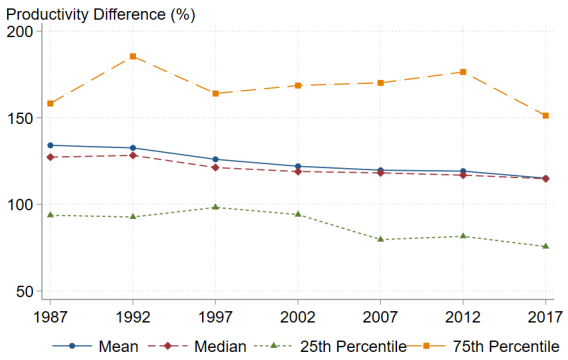
c. Firm and Store 10-1 Ranges

Notes: Graph lines represent the average ranges across industries for stores and firms when using different weighting schemes. Unweighted ranges are unweighted arithmetic averages. Establishment-and firm-weighted ranges use each establishment's or firm's share of hours within that industry. Figures are based on authors' calculations using CRT and BLS data.

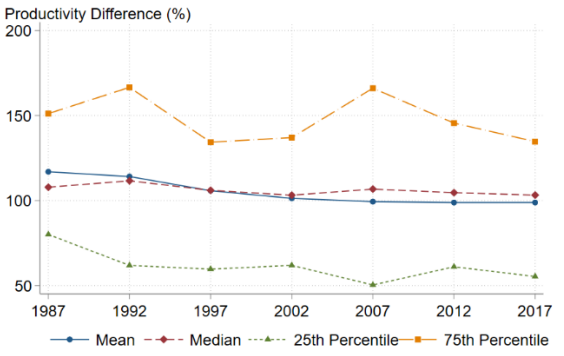
The four 99-90 ranges in panel b have the same ordering as in panel a. But the differences are much smaller, indicating that there are fewer large stores/firms in the upper tail or that highly productive large and small stores/firms have similar productivities. The lower tail differs from the middle and upper tail of the productivity distribution in that the ordering of the IQRs differs, with the weighted 10-1 ranges being higher in some (store-level) or most (firm-level) years. As before, we see significantly more store- and firm-level dispersion in the lower tail than in the upper tail. And, as in the upper tail, there is little difference between the four lines.

Figure 5 compares store- and firm-weighted dispersion statistics. Apart from the differences in mean IQRs (which we saw in Figure 4) and year-to-year variability, store- and firm-based dispersion statistics look remarkably similar. This reinforces the finding in Figure 6 that nearly all of the difference between store- and firm-based dispersion is in the middle half of the productivity distribution.

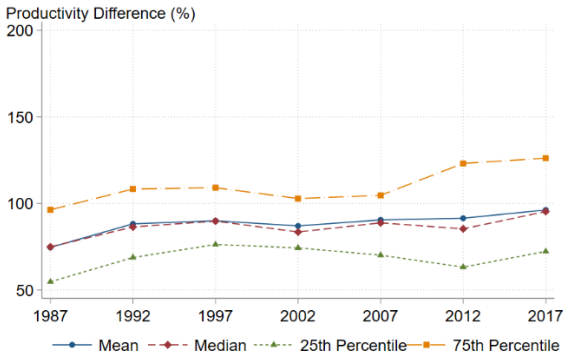
Figure 6: Dispersion across Industries: Store vs. Firm



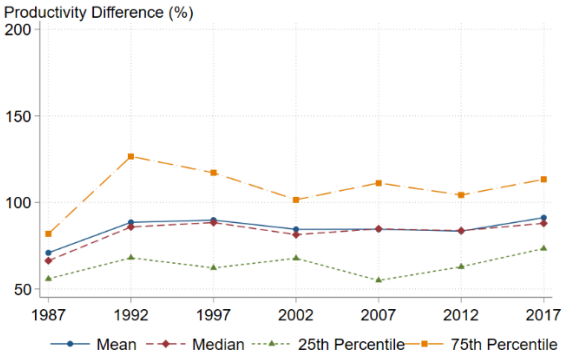
a. Store-weighted IQR



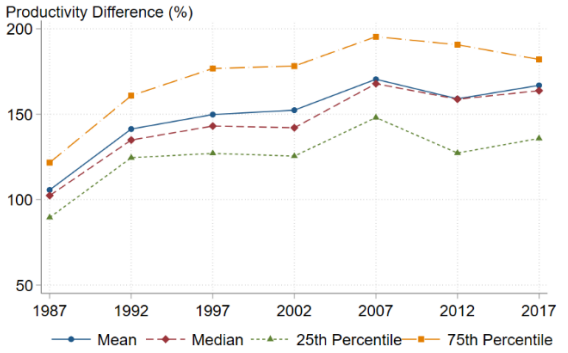
d. Firm-weighted IQR



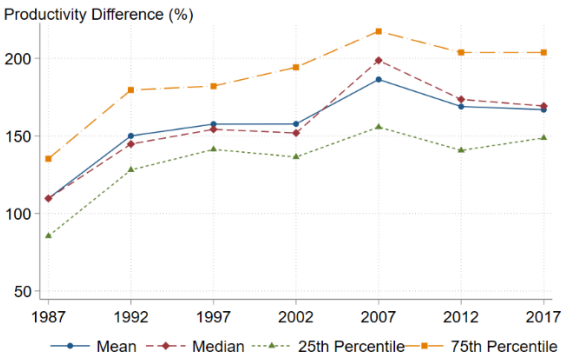
b. Store-weighted 99-90 Range



e. Firm-weighted 99-90 Range



c. Store weighted 10-1 Range



f. Firm-weighted 10-1 Range

Notes to Figures 6a –f: Graph lines represent the variation across industries in the within-industry statistics. The lines indicate the 25th percentile, median, 75th percentile, and mean of the distributions of within-industry ranges across industries. The y-axis is expressed as productivity multiples. Stores and firms are weighted by share of total hours within each industry. Figures are based on authors' calculations using CRT and BLS data.

5. Industry-specific dispersion patterns

Of the 27 4-digit retail industries for each year, we focus on five industries that highlight differences across industries that cannot be seen in the aggregate statistics. These industries were selected because of their relative importance or growth over the last 30 years. The two largest, Grocery Stores (4451) and General Merchandise Stores, including Warehouse Clubs and Supercenters (4523), account for approximately one-fifth of retail sales in 2017. Two other industries, Electronics and Appliance stores (4431) and Electronic Shopping and Mail-Order Houses (4541), have experienced substantial sales growth over the sample period. Finally, we include Clothing Stores (4481) because it has experienced faster productivity growth than most retail industries from 1987 through 2017. Figures 7a-7e show the IQR, the 99-90 range and 10-1 range of labor productivity of establishments for these five industries.

In Figure 7a, the IQR for Grocery Stores (NAICS 4451) was essentially flat over the entire sample period, increasing slightly from about 110 percent in 1987 to 120 in 2017, see Figure 9a. That is, the grocery store at the 75th percentile was 110 percent more productive than a store at the 25th percentile in 1987 and 120 percent more productive in 2017 (that is, 2.1 times and 2.2 times as productive). Dispersion in the tails followed a different pattern. Between 1987 and 2012, dispersion in both tails grew by more than the IQR before falling between 2012 and 2017. The 99-90 difference increased from 50 percent to around 100 percent, with nearly all of the increase occurring between 1987 and 1992. In 1992 and later years, dispersion in the upper tenth of the distribution was about the same as for the middle of the distribution. Dispersion is greater and more variable among the least productive stores (except for 1987 and 1997) compared with the middle half and upper tail of the productivity distribution.

Among General Merchandise Stores (NAICS 4523) in Figure 7b, dispersion levels are similar to Grocery Stores, with dispersion being the greatest in the lower tail of the productivity distribution and about the same for the upper tail and middle half of the

distribution. The IQR declined over the sample period, while dispersion in the tails increased slightly. Like Grocery Stores, dispersion was considerably more variable in the lower tail.

We include Electronic Shopping and Mail-Order Houses (Figure 7c) because this industry saw substantial growth since 1987. There was also a substantial increase in all three dispersion measures. The IQR multiple increased from about 200 percent in 1987 to almost 350 percent in 2017. Some of the dispersion in this industry is possibly due to differences in the product categories that are sold by the different retailers.[9] And given the dramatic growth in industry sales combined with the transformation of this industry from catalog to online sales, it seems likely that some of the increase in dispersion is due to differences in productivity between entering and incumbent establishments. There may also be variation in the tendency of these retailers to outsource sales-related activities such as delivery logistics.

The trends in productivity dispersion in the Electronics and Appliances (NAICS 4431) industry differ from the other retail industries (Figure 8d) in that the IQR declined sharply between 1987 and 2002 and then increased slightly between 2002 and 2017. Dispersion in the tails was lower or the same as the IQR. Like most of the other industries, there is less dispersion in the upper tail than in the other parts of the distribution.

The final industry we consider is Clothing Stores (NAICS 4481). Figure 7e shows that the IQR was relatively stable over the sample period at just over 100 percent, while both the 99-90 and 10-1 percentile ranges increased substantially. Compared with the other four industries, the dispersion ranges for clothing stores exhibited less volatility even though this industry experienced significant productivity growth.

[9] In the 2022 NAICS, e-commerce activities are included with the industry associated with the products sold. For example, online clothing sales will be included in Clothing rather than Electronic Shopping.

Figure 7a: Within-Industry Dispersion – Grocery Stores (4451)

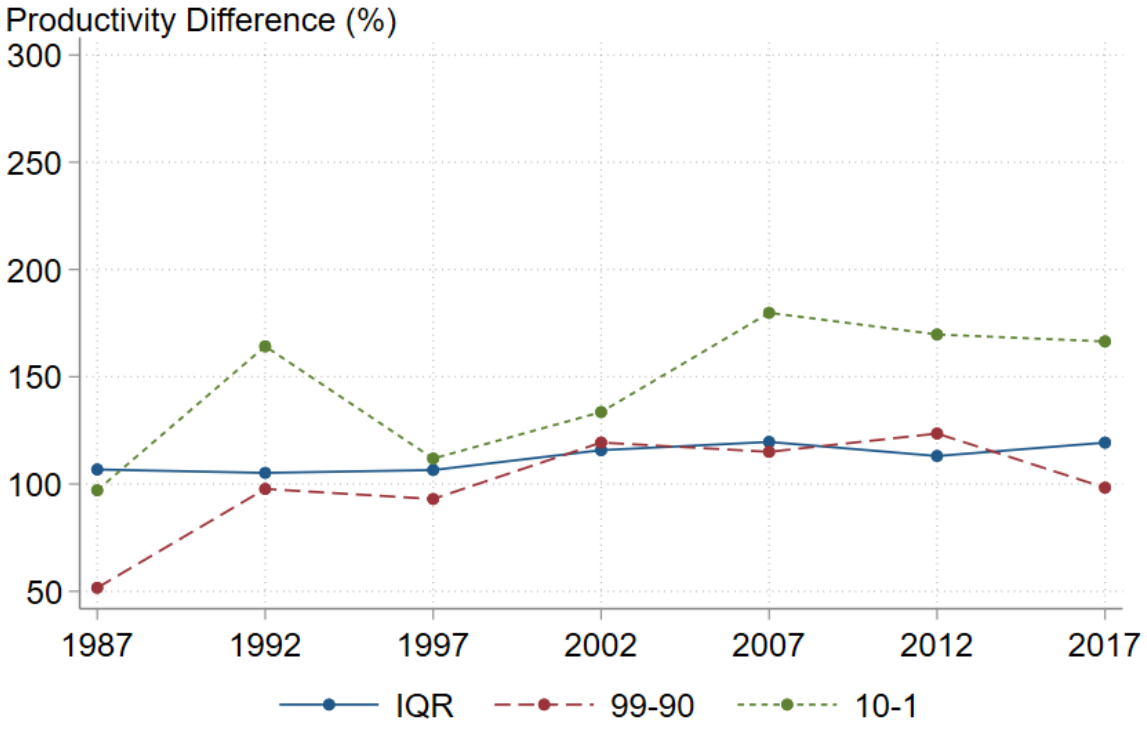


Figure 7b: Within-Industry Dispersion - General Merchandise Stores, including Warehouse Clubs and Supercenters (4523)

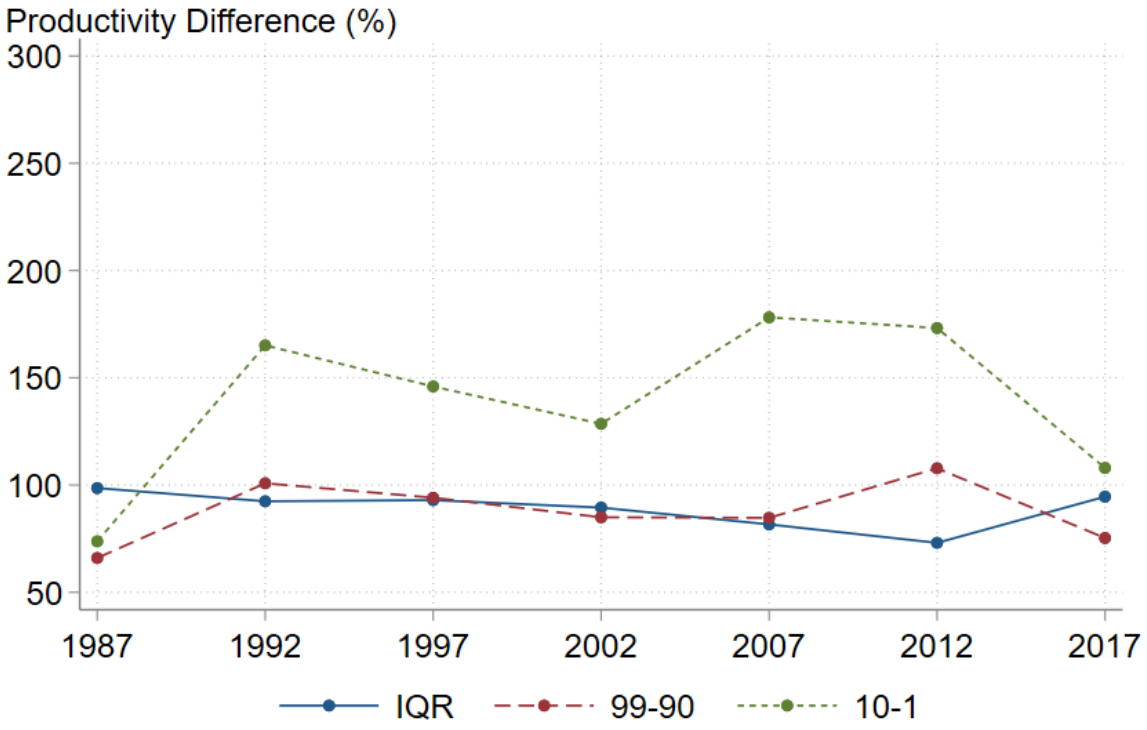


Figure 7c: Within-Industry Dispersion – Electronic Shopping and Mail-Order Houses (4541)

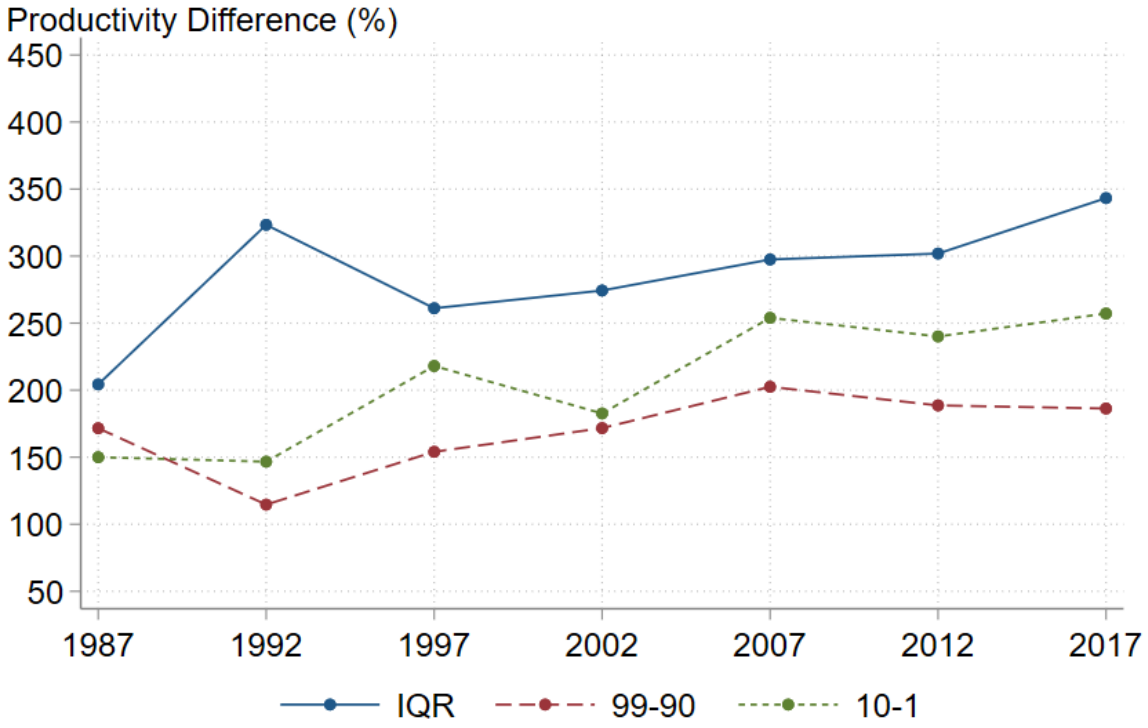


Figure 7d: Within-Industry Dispersion – Electronics and Appliances (4431)

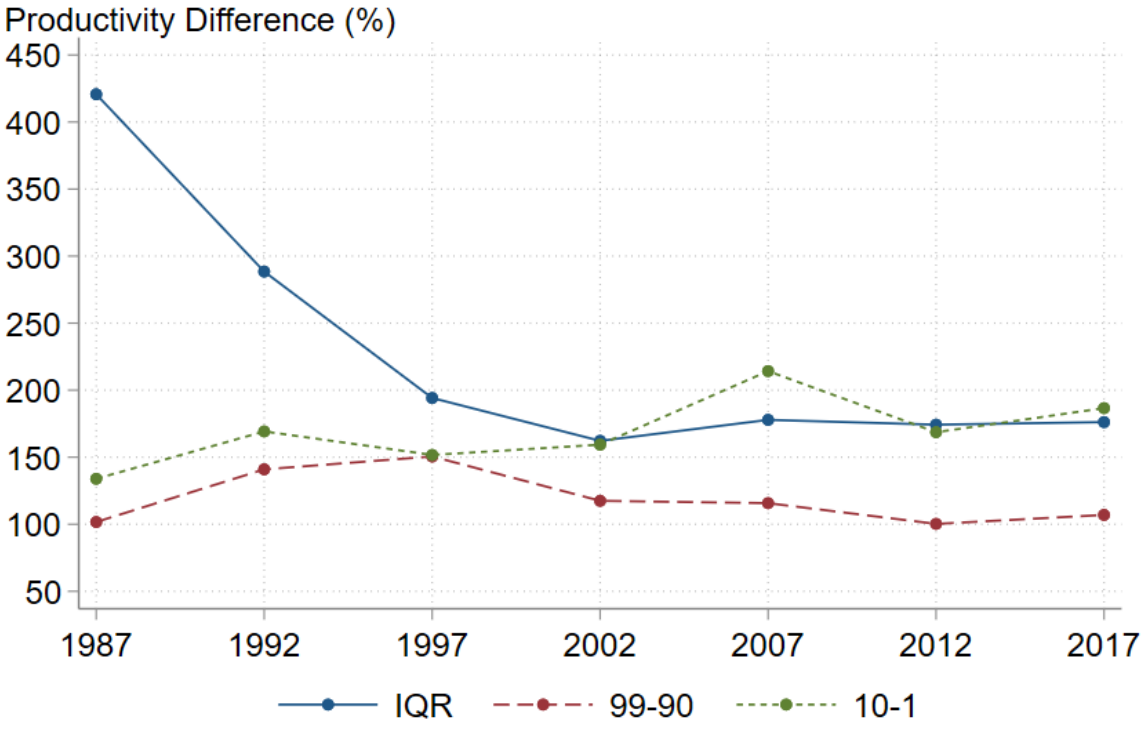
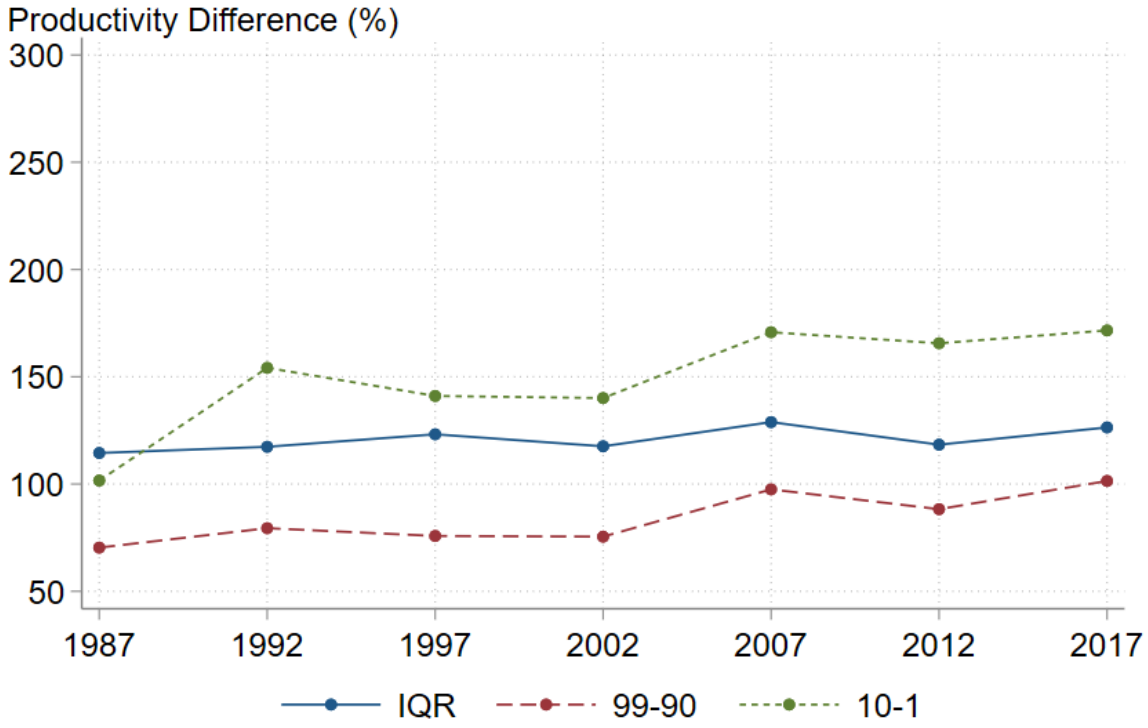


Figure 7e: Within-Industry Dispersion – Clothing Stores (4481)



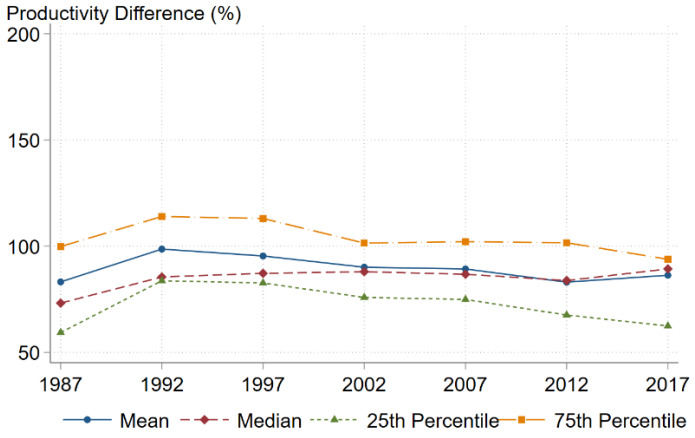
Notes for Figures 9a-9e: The lines represent the unweighted difference between establishments at different percentiles of the within-industry labor productivity distribution in each year. The y-axis is expressed in percentage differences. Based on authors' calculations using CRT and BLS data.

The main takeaway from these figures is that the levels and trends in productivity dispersion can vary considerably across retail industries. For example, grocery stores and general merchandise stores were fairly similar to each other, but levels and trends for e-commerce and electronics & appliance stores were quite different. In some industries there is as much or more dispersion in the upper and lower decile of the distribution as there is in the middle half. In all of these industries, there is more dispersion among stores in the lower tail compared to the upper tail and in some cases the middle of the distribution.

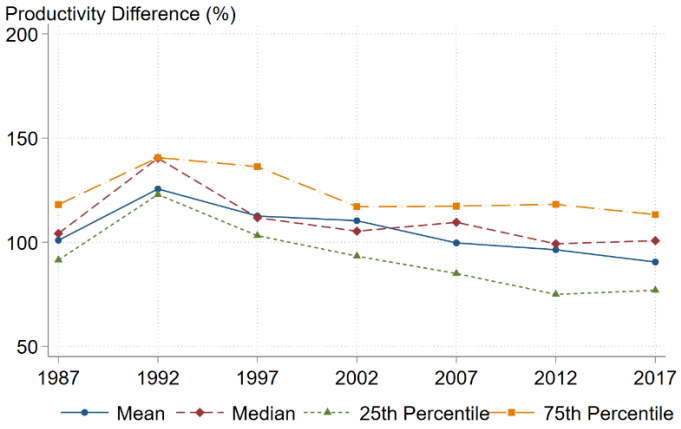
6. Payroll vs. Hours

Our use of industry averages for hours per employee means that the variation in total hours across establishments is determined by variation in total employment and does not account for variation in the use of part-time vs. full-time employees. And hours worked during the March reference period may not be representative of the entire year. As a robustness check, we repeat the dispersion statistics shown in Figure 8 using total payroll in the denominator rather than total hours. It should be noted that payroll also reflects other things besides hours worked such as the mix of occupations (which could also reflect the substitution of other factors for labor), worker-firm effects, leave policies, and any compensating wage differentials, and in this sense this measure can be considered more comprehensive. Figure 10 shows that using payroll implies significant differences. First, using payroll implies considerably less measured dispersion, see Figure 10a: all three lines are below their counterparts in Figure 4a. This indicates that payroll measures may capture additional variation in effective labor, including differences in hours and differences in worker skills and occupation, as discussed in Blackwood, et al. (2024). The second notable difference is that, starting in 1992, payroll-based productivity dispersion decreased (except for the median IQR) whereas hours-based productivity dispersion increased.

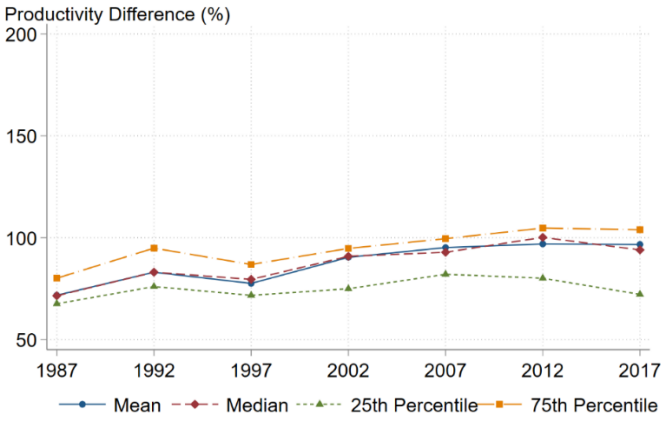
Figure 8: Dispersion across Industries: Establishment-Payroll-Based



a. IQRs



b. 99-90 Ranges



c. 10-1 Ranges

Notes: Graph lines represent the variation across industries in the within-industry statistics. The lines indicate the 25th percentile, median, 75th percentile, and mean within-industry IQR across industries. The y-axis is expressed in log points. Productivity is calculated as the log ratio of sales to payroll, which implicitly assumes the deflator for sales and payroll should be the same. NAICS-4 industries are weighted according to their share of total hours within a year. Establishments within each industry are given equal weight. Figures are based on authors' calculations using CRT and BLS data.

Turning to the 99-90 ranges in Figures 2b and 8b, we see similar differences in time trends. The dispersion measures for payroll-based productivity start at a higher level, but the two sets of lines converge and remain at approximately the same level for the last 10 years of the sample. Finally, looking at the left tails, the lines in both Figures 5c and 10c trend up. The payroll-based productivity dispersion statistics are lower than the hours-based ones, and they exhibit significantly less variation.[10]

7. Comparison to Manufacturing

Although the market conditions facing retail stores differ from those faced by manufacturing plants during this period, the manufacturing sector is a useful comparison, because there has been little research on productivity distribution outside of manufacturing. Figures 9a, b, and c show the percentile ranges (both establishment weighted and unweighted) for manufacturing and retail trade.

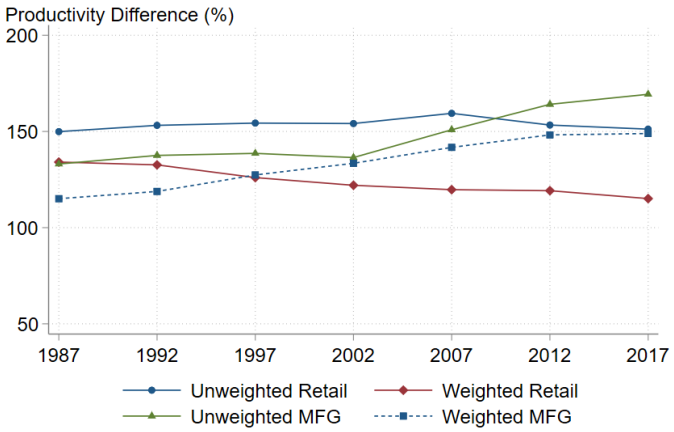
Starting with the IQRs in Figure 9a, we see the familiar pattern that the unweighted IQRs are greater than establishment-weighted IQRs in both sectors. Looking at the unweighted lines, we see that there is more dispersion in retail trade and that the trends in the two sectors are similar until 2002. After 2002, the manufacturing IQR increases rapidly, while the retail IQR remains relatively flat. Turning to the weighted IQRs, we see more significant differences. The weighted retail IQR starts at a higher level but exhibits a strong downward trend, while the weighted manufacturing IQR trends strongly upward.

Turning to the right tail, the two sectors have different levels of dispersion, but they have similar trends. Right-tail dispersion is greater in manufacturing whether or not we weight by hours. However, the difference is smaller when establishments are weighted. Further, the difference between the weighted and unweighted lines is much larger in manufacturing.

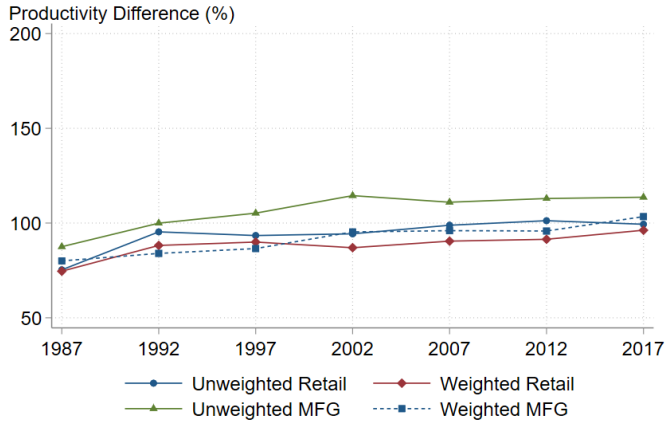
[10] Although the standard deviation is sensitive to outliers, we repeated the analysis using this measure for robustness. The main implication is that dispersion measured by the standard deviations exhibits the same patterns that we documented above.

More investigation into these differences is warranted but is beyond the scope of this paper. We saw above that the shift from single stores to national chains, which open larger and more homogeneous stores. Further, some new technologies such as scanners and self-checkout can be retrofitted into existing stores. In contrast to retail, new plants in manufacturing industries are more likely to be more productive than existing plants because they can more easily incorporate the latest technology, regardless of size. Older plants are less likely to be retrofitted because the existing structure may not be able to accommodate the new technology. Thus, as new manufacturing plants are built, we would expect dispersion to increase. Differences in the left tails of the distributions likely have a different explanation. To speculate, one possibility is that stores in the left tail of the retail distribution may be more likely to survive, relative to manufacturing establishments, if they serve geographically isolated areas where there is little competition or if they cater to niche markets. In manufacturing, new plants are more productive than older and less-productive plants, making those plants less likely to survive.

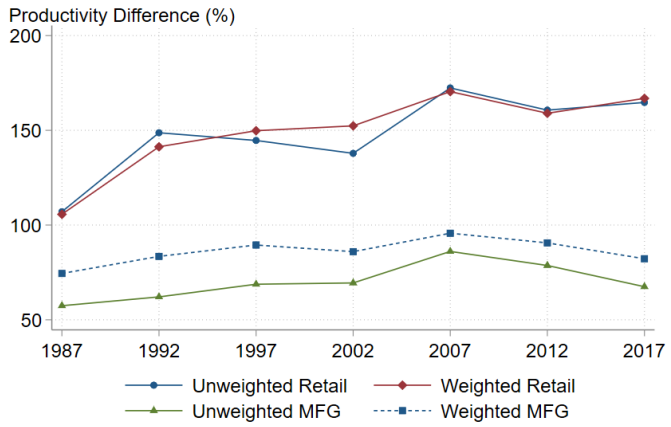
Figure 9: Comparison of Retail Trade and Manufacturing



a. IQRs



b. 99-90 Ranges



c. 10-1 Ranges

Notes to Figures 9a,b,c: Graph lines represent the average ranges (IQR, 99-90, 10-1) across industries when using different weighting schemes. Unweighted is an unweighted arithmetic average. Weighted ranges are calculated using each establishment's share of hours within that industry. Figures are based on authors' calculations using CRT and BLS data.

8. Concluding Remarks

In this paper we present dispersion statistics for retail trade industries. Our work builds on earlier work covering manufacturing industries. We discuss measurement issues and detail the methods by which labor productivity is measured at the retail industry level. We then consider different ways of measuring productivity dispersion. These results shed light on the transformation of the retail trade sector and highlight some of the differences between retail trade and manufacturing.

The retail trade sector has seen dramatic changes since the late 1980s as single stores have been replaced by stores in national chains, which are larger and more homogeneous. This transformation can be seen in the dispersion statistics by looking at the retail sector along different dimensions. Most of the impact of national chains on dispersion shows up in the middle of the productivity distribution, which is consistent with other findings that much of the expansion of national chains was in the middle of the productivity distribution. Comparing hours-weighted to unweighted mean IQRs and store-based to firm-based mean IQRs, we see the impact of the expansion of national chains. Store-based dispersion is higher and exhibits an upward trend with no weighting. But when stores are weighted by hours, the mean IQR has a downward trend. Comparing firm-based and store-based dispersion statistics, a similar story emerges. Unweighted firm-based dispersion is larger than store-based dispersion in levels, although this is reversed when we weight by hours. This reflects the homogeneity and larger size of stores in national chains, which in some sense results in double weighting firms. Interestingly, firm-weighted and store-weighted mean IQRs have similar trends.

Weighting by industry share of total hours worked yields lower dispersion levels but has relatively little effect on trends. We further explore industry-level trends of some of the most important industries and find considerable differences in levels and trends.

Comparing the retail sector to manufacturing, we saw some interesting differences and similarities. In the middle half of the productivity distribution, the main difference is the trend in establishment-based hours-weighted IQRs. The manufacturing IQR exhibits are strong upward trend, while the retail IQR exhibits a strong downward trend. The two sectors are more similar in the right tail of the productivity distribution. The levels are different, with manufacturing having more dispersion, but dispersion in the upper tails trends upward in both sectors. The biggest differences between retail and manufacturing are in the left tail. Left-tail dispersion is much greater and increases more sharply in retail compared with manufacturing. This stark difference could potentially be related to the fact that, despite consolidation, a substantial amount of retail activity is still accounted for by single-units, and these tend to have lower productivity. Furthermore, dispersion in the left tail increased by a similar amount to the right tail over the 30 years we consider. As we speculated, stores in the left tail of the retail distribution may be more likely to survive, relative to manufacturing establishments, because they may serve geographically isolated areas where there is little competition or they may cater to niche markets. In contrast, new manufacturing plants can outcompete older plants, making less productive plants less likely to survive. These findings suggest further exploration of the left tail of the distribution is warranted, which we relegate to future research.

As in manufacturing industries, we find that there is significantly less productivity dispersion when labor input is defined as payroll rather than hours. This could be the result of unobserved factors such as differences in the mix of occupations or worker-firm effects. But it could also represent mismeasurement of hours due to variation in work hours across establishments or the March reference period for employment counts. Thus, accounting for heterogeneity in labor could be important in retail. Future research will further investigate this result and shed additional light on the sources of productivity dispersion.

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Table A.1 – Industry Share of Employee Hours

NAICS	Industry	1987 Share	2017 Share
4451	Grocery stores	0.192	0.171
4523	General Merchandise Stores, including Warehouse Clubs and Supercenters	0.068	0.152
4411	Automobile dealers	0.093	0.101
4441	Building material and supplies dealers	0.072	0.071
4461	Health and personal care stores	0.054	0.065
4481	Clothing stores	0.063	0.063
4471	Gasoline stations	0.090	0.062
4413	Automotive parts, accessories, and tire stores	0.060	0.041
4541	Electronic shopping and mail-order houses	0.008	0.037
4522	Department stores	0.072	0.028
4511	Sporting goods, hobby, and musical instrument stores	0.023	0.025
4431	Electronics and appliance stores	0.019	0.025
4539	Other miscellaneous store retailers	0.011	0.018
4421	Furniture stores	0.019	0.016
4422	Home furnishings stores	0.015	0.015
4482	Shoe stores	0.015	0.013
4533	Used merchandise stores	0.004	0.012
4543	Direct selling establishments	0.016	0.012
4532	Office supplies, stationery, and gift stores	0.018	0.012
4442	Lawn and garden equipment and supplies stores	0.013	0.011
4412	Other motor vehicle dealers	0.009	0.011
4453	Beer, wine, and liquor stores	0.014	0.010
4452	Specialty food stores	0.014	0.009
4483	Jewelry, luggage, and leather goods stores	0.014	0.008
4512	Book stores and news dealers	0.006	0.005
4531	Florists	0.011	0.004
4542	Vending machine operators	0.006	0.003

Notes: Numbers are the share of retail hours of labor inputs accounted for by each industry in 1987 and 2017. Hours are calculated as the product of employment for each establishment in the Economic Census and the estimate of hours worked per employee from the BLS. This measure is then summed to the industry level.